OCTNet: A Deep Learning Platform for Automated Retinal OCT Image Analysis in Rural Healthcare

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Abstract

In rural India, where eye specialists and advanced equipment are scarce, millions risk blindness from retinal diseases like choroidal neovascularization (CNV) and diabetic macular edema (DME). We developed OCTNet, a deep learning platform for automated analysis of Optical Coherence Tomography (OCT) images, to make retinal screening affordable and accessible. Using MobileNetV3Large, trained on 84,495 OCT images from global medical centers, OCTNet classifies retinas into CNV, DME, Drusen, or Normal with 94.76% accuracy and a weighted F1-score of 0.95. Hosted on a Streamlit web app, it runs on low-cost laptops (Intel Core i5, 8GB RAM), processing images in under 3 seconds. This reduces screening time from 40 minutes to 5 minutes, enabling rural doctors to diagnose and refer patients effectively. Inspired by community vision screening approaches, OCTNet offers a scalable solution to prevent blindness in underserved areas.

Keywords: retinal OCT, deep learning, rural healthcare, MobileNetV3, Streamlit

Introduction

Eye diseases like choroidal neovascularization (CNV) and diabetic macular edema (DME) can steal sight silently, especially in India's villages where eye doctors are rare. Over 62 million Indians face vision impairment, and 80% of cases could be prevented with early detection . Optical Coherence Tomography (OCT) scans, which take detailed pictures of the retina, are great for spotting these diseases, but OCT machines are expensive, and specialists are far away. Rural patients often miss out on timely care.

Our project, OCTNet, aims to bring eye screening to villages using a computer program that acts like a robot doctor, analyzing OCT images to find problems fast. We were inspired by a paper, *Enhancing Community Vision Screening* , which used AI for eye checks in Singapore. We adapted this idea for rural India, using a simple laptop and a web app anyone can use. OCTNet checks for four conditions: CNV (leaky blood vessels), DME (swelling from diabetes), Drusen (early signs of age-related macular degeneration), and Normal retinas. It's like solving a puzzle to tell if an eye is healthy or needs help.

As a student, I used free tools like Semantic Scholar to find papers, Paperpal to write clearly, and Research Rabbit to connect ideas, making this project doable during my internship. This paper shares our work, testing OCTNet on 84,495 images and planning how to use it in rural clinics.

Methods

OCTNet is a deep learning platform with a web interface, like a friendly robot that looks at eye scans and gives answers. Figure 1 shows its workflow. We built it using Python, TensorFlow/Keras for the model, and Streamlit for the web app, all running on a basic laptop (Intel Core i5, 8GB RAM). Here's how it works:

- **Dataset**: We used 84,495 high-resolution OCT images from global medical centers, labeled as CNV, DME, Drusen, or Normal . The dataset was split into 76,515 training, 21,861 validation, and 10,934 test images. Expert ophthalmologists verified labels to ensure accuracy.
- **Model**: We used MobileNetV3Large, pretrained on ImageNet, for transfer learning. It

takes images resized to 224x224 pixels (RGB). The final layer is a dense layer with 4 units (softmax) for classifying the four conditions. We used Categorical Crossentropy loss, the Adam optimizer (learning rate 0.0001), and tracked accuracy and F1-score.

- **Training**: We trained for 15 epochs with a batch size of 32 (train/validation) and 64 (test). Images were preprocessed with standard normalization, learned from my Coursera course on image processing. The model has 5.5 million parameters, with 5.49 million trainable.

- **Web App**: The Streamlit app lets users upload an OCT image and get a diagnosis (e.g., "CNV detected") with medical recommendations, like a doctor's note. It's simple enough for rural nurses to use.

We followed for evaluating performance with precision, recall, and F1-scores, ensuring our model is reliable for real-world use.

Workflow of OCTNet, showing image upload, preprocessing, MobileNetV3Large classification, and output with recommendations.

Results

We tested OCTNet on 10,934 OCT images, using a laptop (Intel Core i5, 8GB RAM). The results are like a report card for our robot doctor:

- **Overall Performance**: Achieved 94.76% accuracy, a test loss of 0.1883, and a weighted F1-score of 0.95. The macro-average F1-score was 0.91.

- **Per-Class Metrics**: Table 1 shows precision, recall, and F1-scores for each class. CNV and Normal had the highest F1-scores (0.96, 0.97), while Drusen was lower (0.79) due to fewer images (888 samples).

- **Efficiency**: OCTNet processes an image in under 3 seconds, using less than 4GB RAM, making it affordable for rural clinics. Compared to (95.2% accuracy), our accuracy is slightly lower but practical for low-cost setups.

Per-class performance metrics on the test set (10,934 images).

Class	**Precision**	**Recall**	**F1-score**	**Support**
CNV	0.97	0.95	0.96	3746
DME	0.95	0.88	0.91	1161
Drusen	0.79	0.79	0.79	888
Normal	0.96	0.99	0.97	5139
Weighted Avg			0.95	10934

Figure 2 shows a confusion matrix, helping us see where the model made mistakes, like confusing Drusen with Normal retinas.

Confusion matrix showing classification performance across CNV, DME, Drusen, and Normal classes.

Discussion

OCTNet makes retinal screening possible in rural India, where traditional OCT machines cost lakhs and take 40 minutes per patient. Our platform uses a cheap laptop and a Streamlit app, finishing in 5 minutes, like a quick checkup. It's accurate (94.76%) and gives clear recommendations, like telling a patient with CNV to see a specialist fast. This builds trust, as nurses can show patients what's wrong, similar to 's approach.

However, Drusen detection (F1-score 0.79) needs improvement due to fewer training images. We also tested mostly on a global dataset, so Indian patient data could make OCTNet even better. As a student, I found tools like Semantic Scholar and Research Rabbit super helpful for finding papers, while Paperpal made my writing clearer. These free tools were a lifesaver during my internship.

Future work includes testing OCTNet in rural clinics and adding more Indian OCT images. We also want to make the app work offline, as internet is spotty in villages. OCTNet is a big step toward saving sight in rural India.

Conclusion

OCTNet is a deep learning platform for automated OCT image analysis, designed for rural India's healthcare needs. Using MobileNetV3Large and a Streamlit interface, it achieves 94.76% accuracy and a 0.95 F1-score on 10,934 test images, running on affordable laptops. It detects CNV, DME, Drusen, and Normal retinas, offering quick diagnoses and recommendations. Future work will validate OCTNet in real clinics and improve Drusen detection. We hope OCTNet helps rural doctors prevent blindness, one scan at a time!

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